INTERNSHIP SUMMARY REPORT

Company Name - Elevance Skills

Project Title- Learn to Build Real time Google Play store data analytics – python

Submitted By – Deepali Srivastava

Email - deepalinovember14@gmail.com

INDEX

INTRODUCTION

The internship report details my experience and accomplishments during my data analytics internship, my primary objective was to learn about how to collect data from real-life sources using the Python programming language and understand how to analyse it. I have also learned the value of being able to access data as part of the decision-making process.

The data analytics internship provided the opportunity to apply all of the theory I had learned in my classes to a variety of real-world projects, which allowed me to develop analytical reasoning ability, critical thinking skills, and decision-making capabilities through using data.

I developed and enhanced my knowledge of the importance of how to gather relevant information from very large data volumes and present it effectively with clear visual graphic representations, and I now have a more complete understanding of how the practice of data analysis can be used to detect trends, monitor performance of entities, and support informed decision-making for companies.

BACKGROUND

The internship project's objective was centered on the examination of data relating to the Google Play Store app alongside user reviews about app usage. This data set includes an extensive array of details relative to a mobile application, e.g., error messages, categories, numbers of installations, user rating levels, and so forth.

The various types of datasets listed above are often utilized within the corporate world when determining how well a mobile application is succeeding and the ways in which an end-user interacts with the app.

After working with data from this particular dataset, I came to understand how real-world datasets contain both "unstructured data" (that which has no clear identification), and missing pieces data is found in multiple formats. The primary aim of my project was to "clean" and prepare the dataset so that it could be used for analysis. Following the preparation phase, I performed several analyses to identify trends within the marketplace, to contrast between paid and free applications, and to monitor the level of engagement of users from various categories.

LEARNING OBJECTIVES

The Internship focused on the following major learning objectives:

- Grasping the Full Data Analytics Process: Raw Data to Final Insight.
- Applying Data Cleaning and Pre-processing Methods for Missing & Inconsistent Values.
- Setting Up Data Filtering & Aggregating Based on Business Rules/Events.
- Creating Interactive & Visually Attractive Dashboards using Plotly.
- Understanding Temporal and Conditional Visibility in Dashboards.
- Enhancing Your Software Engineering/Programming Skills and Analytical Ability.

The above objectives provided the direction and foundation for every activity throughout this internship.

ACTIVITIES AND TASKS

For the entire internship, I was assigned with multiple tasks related to analyzing and visualizing data using different libraries of Python; Specifically: Pandas, NumPy, and Plotly. Each task required the following three steps:

1) Data Preparation; 2) Data Analysis; 3) Data Visualization

One example of a task I completed is grouping the data that displayed average ratings and total reviews per category of application in a bar chart format. The information displayed in this chart allowed insight into which categories are attracting users and receiving positive feedback.

When working on a task related to visualizing global installs by category, I created a series of interactive choropleth maps. The choropleth maps gave a geographical view of where applications are popular.

I created dual-axis charts that compared installs with revenue for free vs paid apps. Creating these types of charts provided insight into how pricing methods affect user acquisition and revenue generation.

In order to analyze install growth patterns over time, I created time-series charts. The time-series charts enabled me to observe trends over weeks and months.

I created bubble charts that investigate the correlation of app size, ratings, and installs. This chart displays the relationship between size and quality of an app and how it affects downloads.

Finally, I created stacked area charts to visually represent the total installs across various categories. The stacked area charts are great for comparison of the growth of the different categories over time.

SKILLS AND COMPETENCIES

Participating in this internship has greatly improved my technical and analytical competencies. I was able to work with Python Code in a practical way using Pandas for manipulating data and NumPy for performing numerical operations. Also, through the internship, I became more proficient in visualizing data using the Plotly library.

In addition to the above, I learned how to clean and transform datasets, apply logical reasoning, debug code, and take into consideration the real-world constraints on businesses. I also learned how to write clean, readable, and organized code which is important for developing maintainable and collaborative projects.

FEEDBACK AND EVIDENCE

The internship allowed me to showcase my abilities as I created visualizations and dashboards from the ground-up. In developing these products, I had multiple reviews performed on the work I produced to ensure it met quality assurance standards; this included verifying that all of the work produced was accurate, that it was clear and readable, and that it adhered to required guidelines (such as having specific filtering criteria or being performed at a particular point in time).

In addition to utilizing feedback during my internship to refine my problem-solving approach as well as refine my structure for organizing my coding, I also learned about error handling through feedback I received during the duration of my internship. I continued to have evaluations done throughout the entire internship, allowing me to constantly develop my performance.

CHALLENGES AND SOLUTIONS

While doing my internship I encountered a number of challenges. The first major challenge was dealing with missing values and inconsistent data which needed to be clean up thoroughly and validated before proceeding with analysis. Also had to deal with errors associated with using the wrong column headers and grouping logics for analysis.

The second major challenge was implementing strict filter conditions in combination with time-series visibility of dashboard items. To do this I tested the code one piece at a time, reviewed intermediate results, and referred back to the appropriate documentation as necessary in order to resolve these issues through a systematic approach.

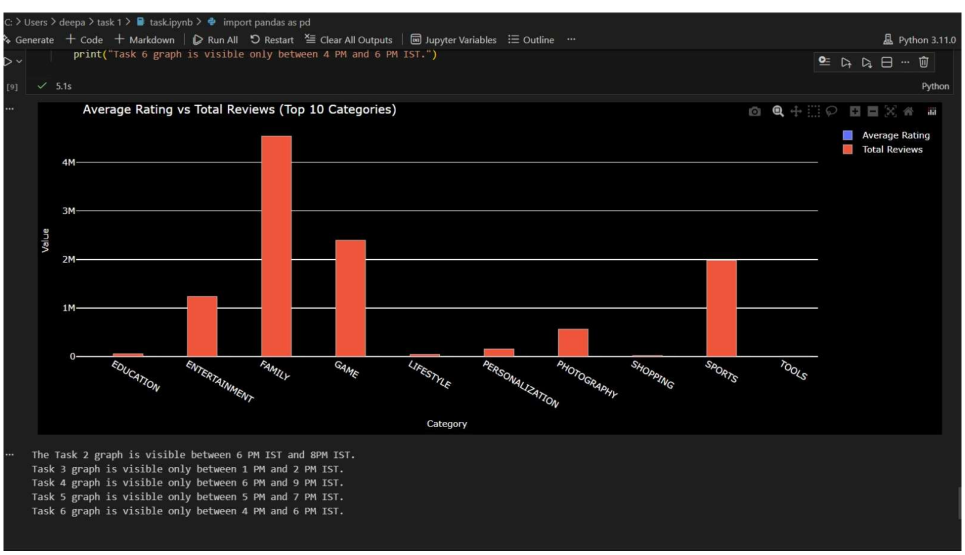
TASK-WISE DOCUMENTATION

Task 1: Grouped Bar Chart – Average Rating vs Total Reviews

Title: Average Rating vs Total Reviews (Top 10 Categories)

Description:
The data in this chart demonstrates the comparative performance of various types of apps within the top 10 most downloaded apps based on number of downloads. There are several factors taken into consideration when determining which apps should be included—apps that have a minimum rating of 4, which are large enough to be viewed at the size specified (at least 10 MB), and all apps that have had an update in the current month of January 2018. This bar chart makes it easy to see how well certain categories are doing when it comes to both quality and user interaction. There is a time-based filter set up in the chart, so it will only display the data during a specified time frame of 3 PM and 5 PM China Standard Time.

Output –



Task 2: Choropleth Map – Global Installs by Category

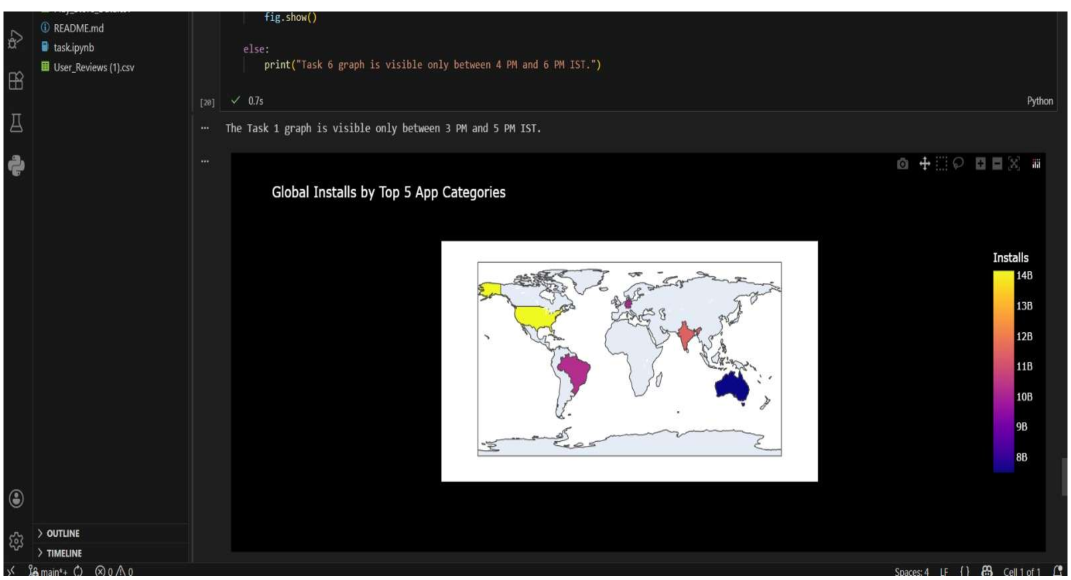
Title: Global Installs by Top 5 App Categories

Description:
As a data analyst intern, this internship provided me the opportunity to experience firsthand all of the fantastic things I learned through my education while utilizing different types of software applications to analyze and predict data and its future trends. As a result, my internship provided an incredible amount of information and knowledge regarding how to integrate everything I learned while attending college.

With this experience and knowledge, I received from my internship, I am confident that I am ready to continue moving up in the field of data analytics, as well as any opportunity of a similar nature.

I believe that my internship experience has been beneficial to my academic and professional development.

Output –

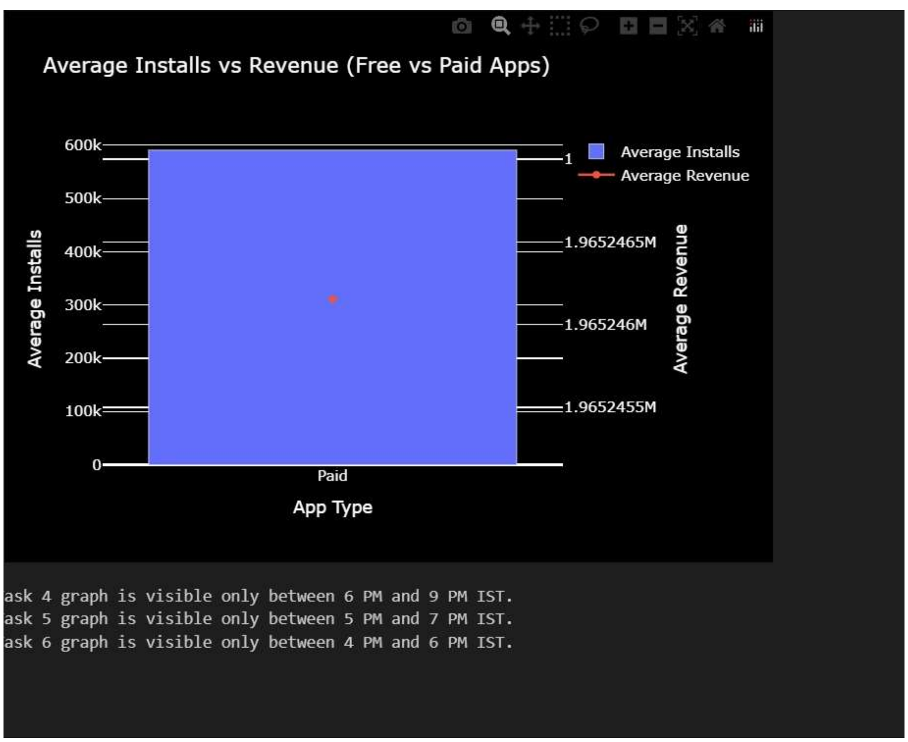


Task 3: Dual Axis Chart – Installs vs Revenue

Title: Average Installs vs Revenue (Free vs Paid Apps)

Description:
This dual-axis chart presents both average installation counts (shown by bar chart) and average revenue generated (shown with a line chart) for the same collection of mobile applications. Each data point within this group was derived from an analysis of various criteria related to installation count, revenue, operating system compatibility (Android Version), size of the application, rating (content), and length of the application name. This chart reveals an obvious difference in how much money has been generated through monetizing free versus paid applications, making it a very useful tool for understanding various strategies used to monetize applications. The live view of this dashboard is available only after the hours of 1pm - 2pm IST.

Output –



Task 4: Time Series Analysis – Category-wise Installs

Title: Total Installs Trend Over Time (Category Wise)

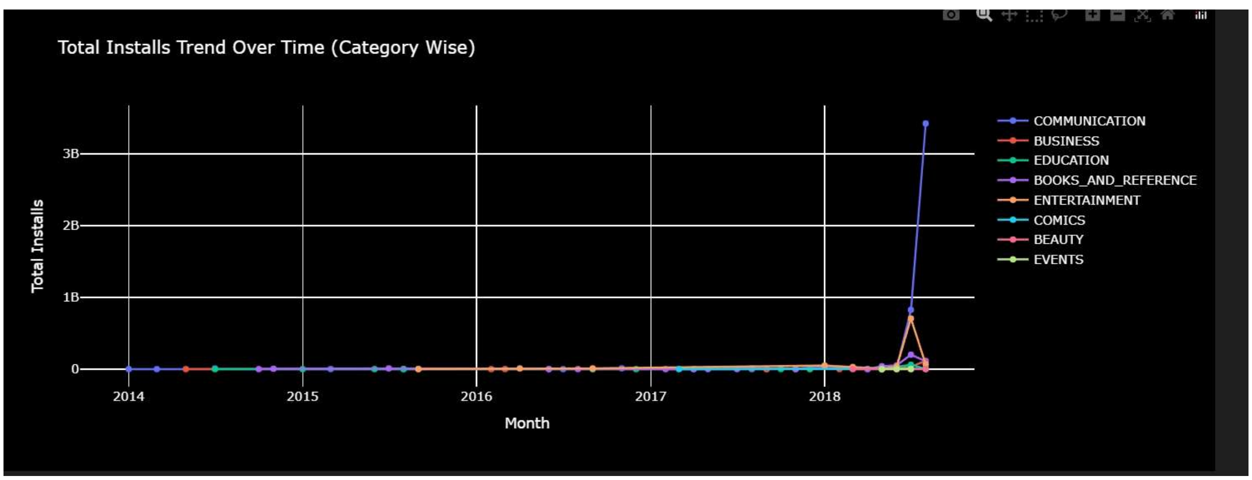
Description:
The example below demonstrates how to view the Installation trends by Category as a time series visualization of Monthly through advanced techniques for filtering based on Review Thresholds and string based.

Category names were translated into several languages to show support for localizing.

Evaluating how trends change by Month over Month Growth can give insight into category performance being affected by different types of changes.

This visualization is available to view from 18:00 – 21:00 IST.

Output –

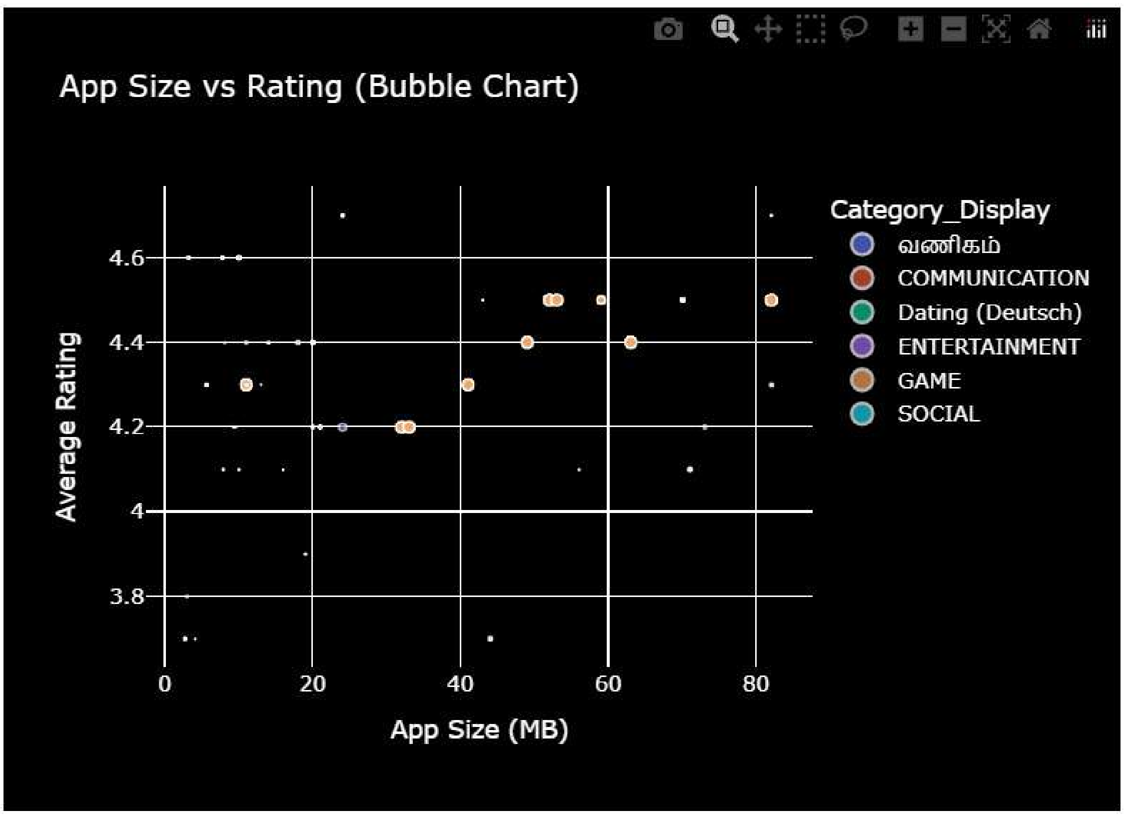


Task 5: Bubble Chart – App Size vs Rating

Title: App Size vs Rating (Bubble Chart)

Description:
In this analysis of the correlation between size and ratio of the application's average score, the size of the bubble represents the volume of installs. In addition to this correlation, there were strict filter restrictions on each application based on the following: Average Score, Reviews, Installs Count, Sentiment Subjectivity, and Category selection. However, translated labels make categories read better for users around the world. By visualizing the data, you can see which ranges provide the best application scores for users. This chart can be viewed anytime between 5 PM - 7 PM IST.

Output –

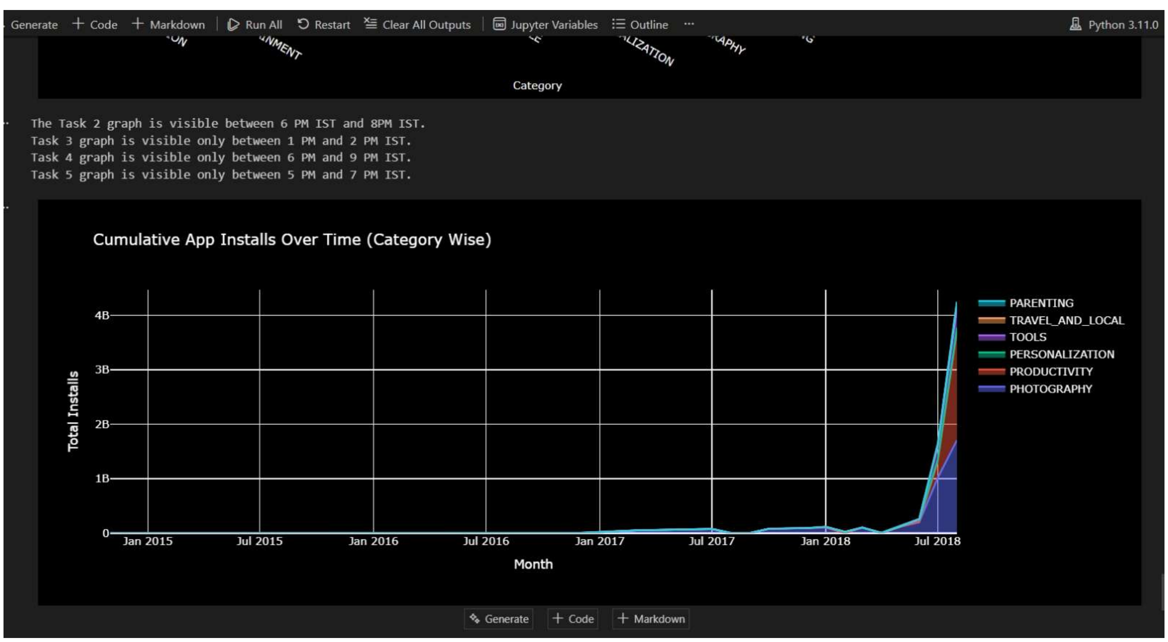


Task 6: Stacked Area Chart – Cumulative Installs

Title: Cumulative App Installs Over Time (Category Wise)

Description:
This stacked area chart displays the cumulative trend for installations in each category over time. Filters were applied to the Categories based on Quality and Size before doing this analysis, allowing for an accurate comparison of similar types of service Providers across the globe. Since Categories operate in multiple Languages, Category names appear on the chart in a variety of Languages to include localized versions, and cumulative Monthly Growth as well as Monthly Over Month (MoM) provide insight into Category Performance on a long-term basis. The chart is available from 4:00 PM to 6:00 PM Indian Standard Time (IST).

Output –



OUTCOMES AND IMPACT

I became an independent user of Data Analytics techniques and completed the development of interactive dashboards during my internship. As a result of my internship experience, I have built up my confidence in working with real-life datasets, further developed my knowledge of how to apply Data Analytics principles and processes in real-life situations, and improved my capacity to derive useful and practical insights from raw datasets to support decision-making. The skill sets developed during this internship will also be of assistance in my future academic research as well as in future employment opportunities related to Data Analytics.

CONCLUSION

In conclusion, this was a wonderful experience for myself as a data analyst intern. This opportunity allowed me to apply the interesting things I have learned in my educational program as well as to apply various kinds of software applications to evaluate data and make predictions regarding future trends. This internship provided immense insight and knowledge on how to apply everything I have learned over the course of my university studies.

Overall, I've gathered enough knowledge from my internship and feel prepared to continue to advance my career in the data analytics world, and any initiatives that are similar.

Overall, my internship experience has contributed positively to my growth in both my academic and professional life.